EXAMINATION RESULT AND GRACE-MARK VALIDATION

Python Project Report

1.Overview

The Examination Result and Grace-Mark Validation System is designed to automate the process of calculating examination results and validating grace marks for students. The system checks students' marks against predefined examination rules and eligibility criteria. It identifies students who are eligible for grace marks and calculates their final results according to the applicable institutional policies.

The main objective of this system is to reduce manual errors, improve accuracy, save time, and ensure transparency in examination result processing. It can generate the final results based on students' marks, passing criteria, and authorized grace-mark policies.

2.Example Policy Used

For demonstration purposes, the following sample policy is used. It is an illustrative policy and does not represent the official rules of any particular institution.

1. The minimum passing mark for each subject is 40 out of 100.
2. A student who scores at least 35 marks but less than 40 marks in a subject may be considered for grace marks.
3. A maximum of 5 grace marks may be awarded in one subject.
4. Grace marks can be awarded only if the student satisfies the institution's eligibility criteria.
5. The final marks must not exceed the maximum marks allotted for the subject.
6. Students who do not qualify for grace marks must retain their original marks.
7. The system must record the original marks, grace marks awarded, and final marks separately.
8. All grace-mark decisions must comply with the institution's approved examination policy.

Example:

- Student's original marks: 37
- Minimum passing marks: 40
- Required grace marks: 3
- Maximum permitted grace marks: 5

- Final marks: 40
- Result: Pass, provided all other applicable examination rules are satisfied.

3.Problem Statement

1. Manual examination result processing is time-consuming and may lead to calculation errors.
2. Identifying students eligible for grace marks requires checking multiple eligibility conditions.
3. Incorrect application of grace-mark rules may lead to inaccurate examination results.
4. Maintaining original marks and grace-adjusted marks separately can be difficult when handled manually.
5. There is a need for a reliable system to validate marks according to predefined examination policies.
6. Generating accurate and transparent final examination results is challenging when the number of students is large.

The proposed system addresses these problems by automating result calculation and grace-mark validation using predefined rules.

4.Objectives

1. To automate the examination result calculation process.
2. To validate student marks according to predefined passing criteria.
3. To identify students eligible for grace marks.
4. To apply grace marks according to the approved examination policy.
5. To minimize human errors in result processing.
6. To maintain accurate records of original marks, grace marks, and final marks.
7. To generate reliable and transparent examination results.
8. To reduce the time required to process examination results.
9. To ensure that grace marks are awarded only to eligible students.
10. To generate the final result status, such as Pass or Fail.

5.Assumptions

1. Student identification numbers are unique.

2. The marks entered into the system are accurate and verified.
3. The maximum marks and minimum passing marks are predefined.
4. The grace-mark policy is approved by the concerned institution.
5. The system receives valid student examination records.
6. Grace marks are awarded only when all eligibility conditions are satisfied.
7. The system does not modify the original marks entered for a student.
8. Final marks are calculated according to the applicable examination rules.
9. Missing or invalid marks are flagged for verification.
10. The final result is generated only after the validation process is completed.

6.Algorithm

Algorithm 1: Examination Result Validation

Step 1: Start.

Step 2: Read the student ID and examination marks.

Step 3: Validate the entered marks.

Step 4: Check whether the marks are within the permitted range.

Step 5: Compare the marks with the minimum passing marks.

Step 6: If the student meets the passing criteria, assign the result as Pass.

Step 7: Otherwise, check whether the student is eligible for grace marks.

Step 8: If the student is eligible, execute the grace-mark validation algorithm.

Step 9: If the student is not eligible, assign the result as Fail.

Step 10: Store the validated result.

Step 11: Display the final result.

Step 12: Stop.

Algorithm 2: Grace-Mark Validation

Step 1: Start.

Step 2: Read the student's original marks.

Step 3: Read the minimum passing marks.

Step 4: Calculate the marks required to pass.

Required Grace Marks = Minimum Passing Marks − Original Marks

Step 5: Check whether the required grace marks are greater than zero.

Step 6: Check whether the student satisfies all grace-mark eligibility conditions.

Step 7: Check whether the required grace marks are within the maximum permitted limit.

Step 8: If all conditions are satisfied, award the required grace marks.

Step 9: Calculate the final marks.

Final Marks = Original Marks + Grace Marks Awarded

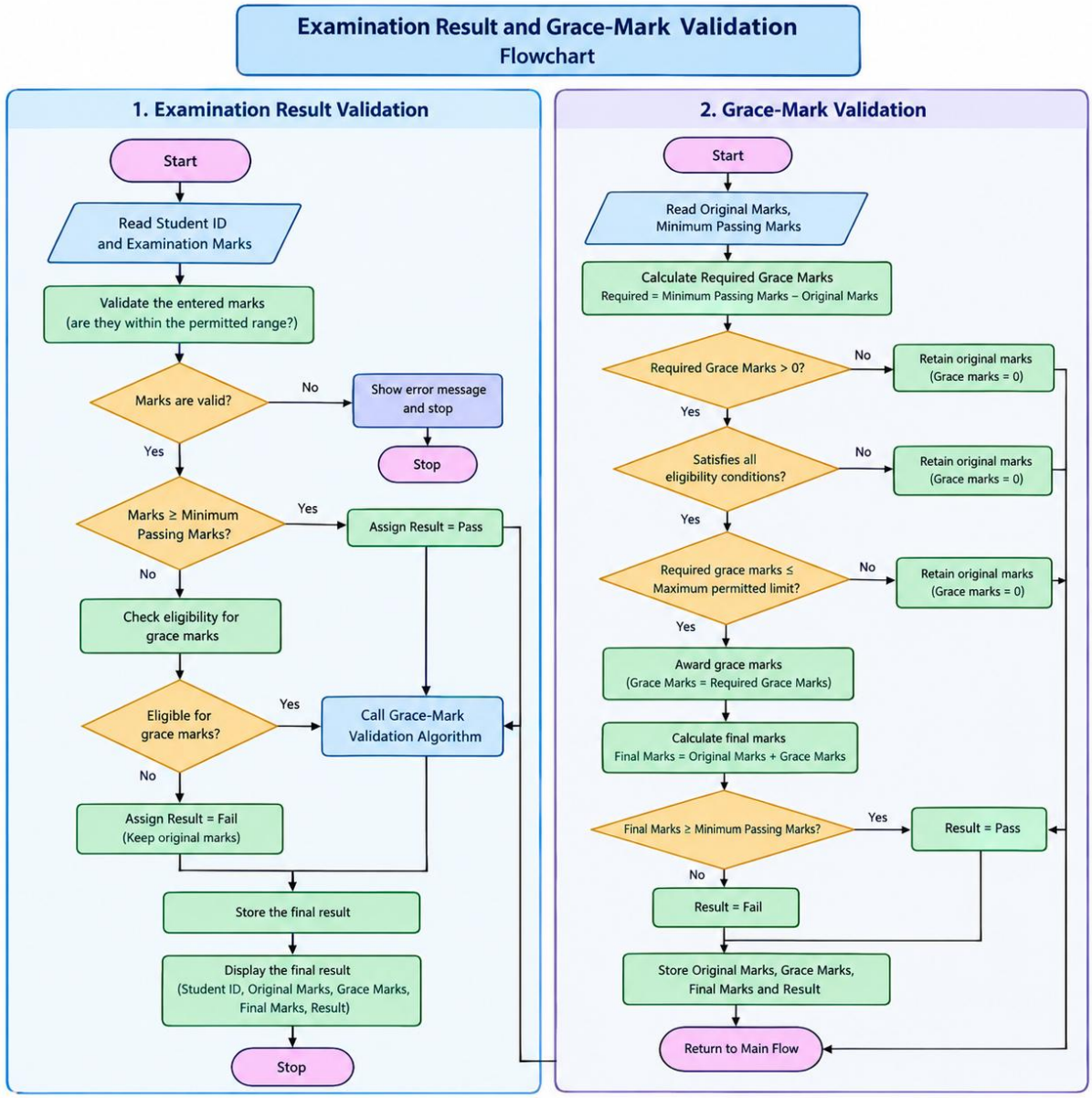
Step 10: Determine the result based on the applicable examination rules.

Step 11: If the student is not eligible, retain the original marks and assign the appropriate result status.

Step 12: Store the original marks, grace marks awarded, final marks, and result status.

Step 13: Stop.

7.Flowchart



8.Pseudocode

START

INPUT Student_ID, Marks

IF Marks < 0 OR Marks > 100

  PRINT "Invalid Marks"

ELSE IF Marks >= 40

  Result = "PASS"

ELSE

  Grace = 40 - Marks

```
    IF Grace <= 5 AND Eligible = TRUE
        Marks = Marks + Grace
        Result = "PASS"
    ELSE
        Result = "FAIL"
    ENDIF
ENDIF
PRINT Student_ID, Marks, Result
STOP
```

9.Source Code

```
# Examination Result and Grace-Mark Validation

print("EXAMINATION RESULT AND GRACE-MARK VALIDATION")
print("-------------------------------------------")

student_id = input("Enter Student ID: ")
marks = int(input("Enter Marks (0-100): "))

PASS_MARK = 40
MAX_GRACE = 5

if marks < 0 or marks > 100:
    print("Invalid Marks! Enter marks between 0 and 100.")

elif marks >= PASS_MARK:
    print("\nOriginal Marks:", marks)
    print("Grace Marks: 0")
    print("Final Marks:", marks)
```

```
    print("Result: PASS")

else:
    grace = PASS_MARK - marks

    if grace <= MAX_GRACE:
        final_marks = marks + grace

        print("\nOriginal Marks:", marks)
        print("Grace Marks:", grace)
        print("Final Marks:", final_marks)
        print("Result: PASS (With Grace Marks)")

    else:
        print("\nOriginal Marks:", marks)
        print("Grace Marks: 0")
        print("Final Marks:", marks)
        print("Result: FAIL")
```

10.Test Cases

Test Case ID	Student ID	Input Marks	Expected Grace Marks	Expected Output
TC01	S101	80	0	PASS
TC02	S102	40	0	PASS
TC03	S103	38	2	PASS (With Grace Marks)
TC04	S104	35	5	PASS (With Grace Marks)
TC05	S105	34	0	FAIL

Test Case ID	Student ID	Input Marks	Expected Grace Marks	Expected Output
TC06	S106	0	0	FAIL
TC07	S107	100	0	PASS
TC08	S108	-5	—	Invalid Marks
TC09	S109	105	—	Invalid Marks

11.Output Screenshots

```
EXAMINATION RESULT AND GRACE-MARK VALIDATION
-------------------------------------------
Enter Student ID: S101
Enter Marks (0-100): 45

Original Marks: 45
Grace Marks: 0
Final Marks: 45
Result: PASS

================ RESTART: C:\Users\VARSHINI\Desktop\python\activity2.py
EXAMINATION RESULT AND GRACE-MARK VALIDATION
-------------------------------------------
Enter Student ID: S102
Enter Marks (0-100): 38

Original Marks: 38
Grace Marks: 2
Final Marks: 40
Result: PASS (With Grace Marks)

================ RESTART: C:\Users\VARSHINI\Desktop\python\activity2.py
EXAMINATION RESULT AND GRACE-MARK VALIDATION
-------------------------------------------
Enter Student ID: S103
Enter Marks (0-100): 30

Original Marks: 30
Grace Marks: 0
Final Marks: 30
Result: FAIL
```

12.Conclusion

The Examination Result and Grace-Mark Validation system helps to calculate examination results accurately and efficiently. It validates student marks, checks eligibility for grace marks, and determines the final result based on predefined rules. This system reduces manual errors, saves time, and improves the transparency of result processing. It also ensures that grace marks are awarded only to eligible students

according to the specified policy. Thus, the system provides a simple and reliable solution for examination result validation.